

```
frm_charzp.Picture1.Scale (-10, 0)-(-0, -0.01)
For I = 10 To 2 Step -2
frm_charzp.Picture1.Line (-I, 0)-(-I, -6), QBColor(14)
frm_charzp.Picture1.Line (-I, 0)-(-I, -0.0001)
frm_charzp.Picture1.Print -I
Next I

For I = 0.01 To 0.001 Step -0.002
frm_charzp.Picture1.Line (0, -I)-(-10, -I), QBColor(14)
frm_charzp.Picture1.Line (0, -I)-(-0.5, -I)
frm_charzp.Picture1.Print -I * 1000 'a
"ma"
Next I

'Calling function to draw
'characteristics

If COMBO1 = "0" Then
zener = Array(100 * 10 ^ (-6), 3.3, 76 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)
IZMAX = zener(2)
D = DISPLAY
Else
If COMBO1 = "1" Then
zener = Array(100 * 10 ^ (-6), 3.6, 69 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)
IZMAX = zener(2)
D = DISPLAY
Else
If COMBO1 = "2" Then
zener = Array(50 * 10 ^ (-6), 3.9, 64 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)

IZMAX = zener(2)
D = DISPLAY
Else
If COMBO1 = "3" Then
zener = Array(10 * 10 ^ (-6), 4.3, 58 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)
IZMAX = zener(2)
D = DISPLAY
Else
If COMBO1 = "4" Then
zener = Array(10 * 10 ^ (-6), 4.7, 53 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)
IZMAX = zener(2)
D = DISPLAY
Else
If COMBO1 = "5" Then
zener = Array(10 * 10 ^ (-6), 5.1, 49 * 10 ^ (-3))
I0 = zener(0)
Vz = zener(1)
IZMAX = zener(2)
D = DISPLAY
End If
End If
End If
End If
End If

'In display function input voltage
'is varied from 0 to Vz in steps of
'.001 and current is 'calculated using
'equation

I = - I0 @ (-V / 2*VT - 1)
VT = 26 mV

'When voltage reaches to Vz, current is
'increased in steps of 0.00001.

Public Function DISPLAY()
For V = 0 To Vz Step 0.001

I = -I0 * (Exp(-V / (2 * 26 * 10 ^ (-3))) - 1)
frm_charzp.Picture1.PSet (-V, I), QBColor(4)
I1 = -I
Next V

For I = I1 To IZMAX Step 0.00001
V = Vz
frm_charzp.Picture1.PSet (-V, -I), QBColor(4)
Next I

V = 7.5
I = 0.005
frm_charzp.Picture1.PSet (-V, -I)
frm_charzp.Picture1.Print " Vz = "; Vz;
" V ----> "
End Function

'For help click 'help' button, help form
'will be displayed.

Private Sub Cmd_help_Click()
HELP.Visible = True
HELP.Height = 7000
HELP.Width = 7000
End Sub

'To exit from program 'exit' button is
'used.

Private Sub cmdClose_Click()
frm_setupzp.Visible = False
End Sub

'To return to setup window close button
'is used.

Private Sub cmdClose_Click()
frm_charzp.Visible = False
frm_setupzp.Visible = True
frm_charzp.Cls
End Sub

Private Sub cmdExit_Click()
End
End Sub
```

FRM_CHARZP.FRM

```
'Program for characteristics form (frm_
charzp)

'To show reverse characteristics of
'zener diode, frm characteristics is
'used. Height, width 'and position of
'form is decided when form is loaded.
'Grid is also drawn on form when it is
'loaded.

Private Sub Form_Load()
frm_charzp.Height = Screen.Height
frm_charzp.Width = Screen.Width
frm_charzp.Top = 0
frm_charzp.Left = 0

Picture1.Scale (-10, 0)-(-0, -0.01)

For I = 10 To 2 Step -2
Picture1.Line (-I, 0)-(-I, -1), QBColor(6)
Picture1.Line (-I, 0)-(-I, -0.0001)
Picture1.Print -I
Next I
End Sub

'To locate coordinates of any point on
'characteristics following function is
'used.

Private Sub Picture1_MouseDown(Button As Integer, Shift As Integer, X As Single, Y As Single)
Picture1.CurrentX = X
Picture1.CurrentY = Y

Text1.Text = X & " V"
Text2.Text = Y * 1000 & " A"
End Sub

'To return to setup window close button
'is used.

Private Sub cmdClose_Click()
frm_charzp.Visible = False
frm_setupzp.Visible = True
frm_charzp.Cls
End Sub

Private Sub cmdExit_Click()
End
End Sub
```

HELP.FRM

```
'Program for help form (help)
'OLE1 is used in this form to write
'procedure.

Private Sub Form_Load()

OLE1.DataChanged = False
End Sub

'To close help window help button is
'used.

Private Sub CLOSE_Click()
HELP.Visible = False
End Sub
```